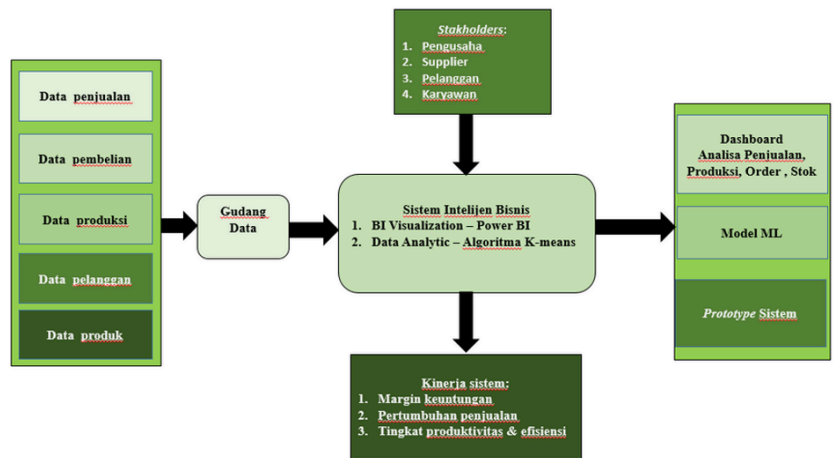
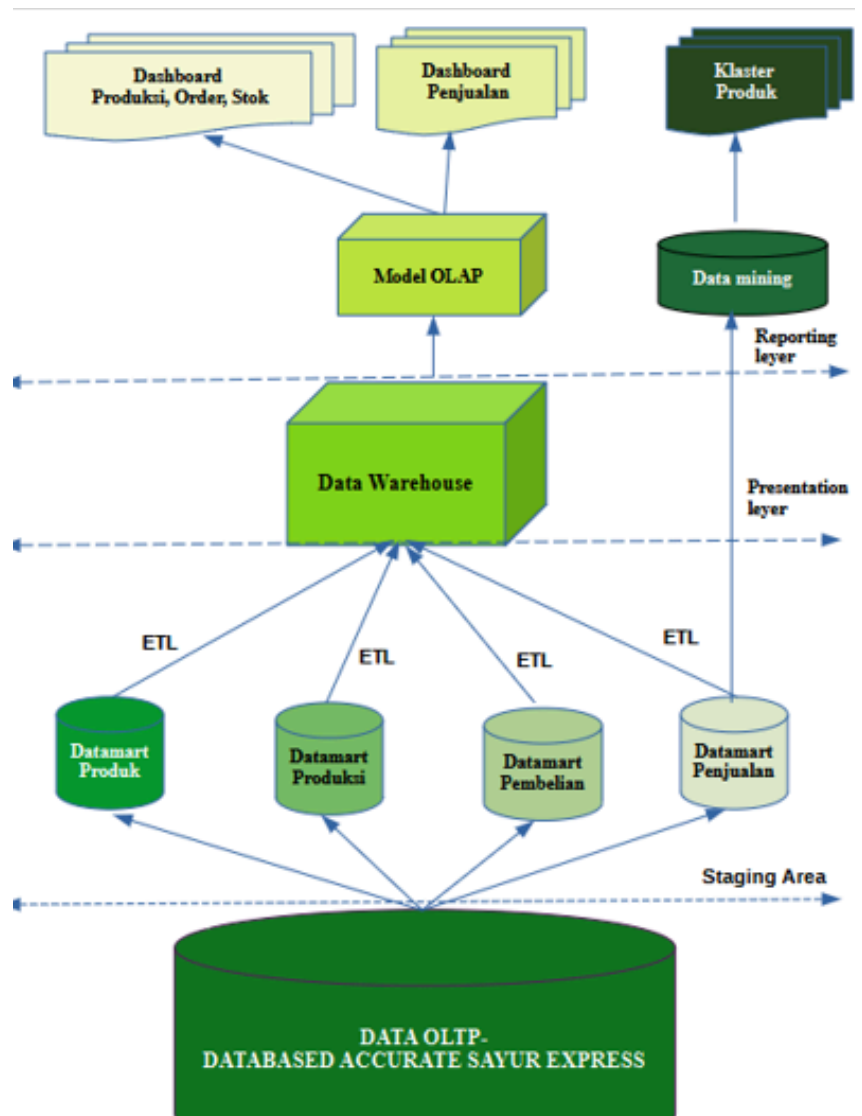


1. Design model sistem intelijen bisnis yang dikembangkan



2. Arsitektur data pipeline



3. Source data Online Order dari API marketplace

Tabel Order Online

#	Name	Type
1	Order_ID 📄	varchar(15)
2	Order_Number	varchar(15)
3	Product_Name	varchar(100)
4	Category_Code	varchar(10)
5	Product_ID 📄	varchar(15)
6	Order_Status	varchar(15)
7	Pick_up_Status	varchar(15)
8	Date_of_Make_Order	date
9	Product_Price	decimal(10,2)
10	Discount_Price	decimal(10,2)
11	Number	decimal(10,2)
12	Returned_quantity	decimal(10,2)
13	Total_Product_Price	decimal(10,2)
14	Discount_Total	decimal(10,2)
15	Discount_by_Seller	decimal(10,2)
16	Discount_by_Shopee	decimal(10,2)
17	Order_Quantity	decimal(10,2)
18	Voucher_by_Seller	decimal(10,2)
19	Cashback_Coin	decimal(10,2)
20	Voucher_by_Shopee	decimal(10,2)
21	Discount_Package	decimal(10,2)
22	Discount_Package_by_Shopee	decimal(10,2)
23	Discount_Package_by_Seller	decimal(10,2)
24	Discount_Coin_by_Shopee	decimal(10,2)
25	Discount_by_Credit_Card	decimal(10,2)
26	Delivery_Cost_by_Seller	decimal(10,2)
27	Discount_of_Delivery_Cost	decimal(10,2)
28	Total_Payment	decimal(10,2)
29	Estimation_Delivery_Cost	decimal(10,2)
30	Product_Size	decimal(10,2)
31	Total_Product_Siza	decimal(10,2)
32	Unit	varchar(5)
33	Customer_ID 📄	varchar(10)
34	Code_Branch	varchar(10)

4. Data understanding dan pre-processing dengan GoogleColab & Rapidminer

Data Shape

```
[12] 1 df_Order = pd.read_excel('/content/Online Order 1-30 Mei 2020.xlsx')
```

```
1 df_Order.shape
```

```
(72533, 50)
```

Seleksi Data--> Finished & On Delivery --- > SGP & SGR

Order Status	count
Finished	63005
Canceled	6605
On Delivery	2694
Needs to be Sent	229

```
[15] 1 df_sales = pd.read_excel('/content/df_order2.xlsx')
```

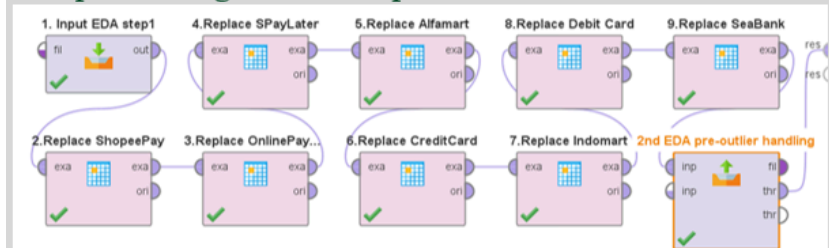
```
1 df_sales.shape
```

```
(40916, 12)
```

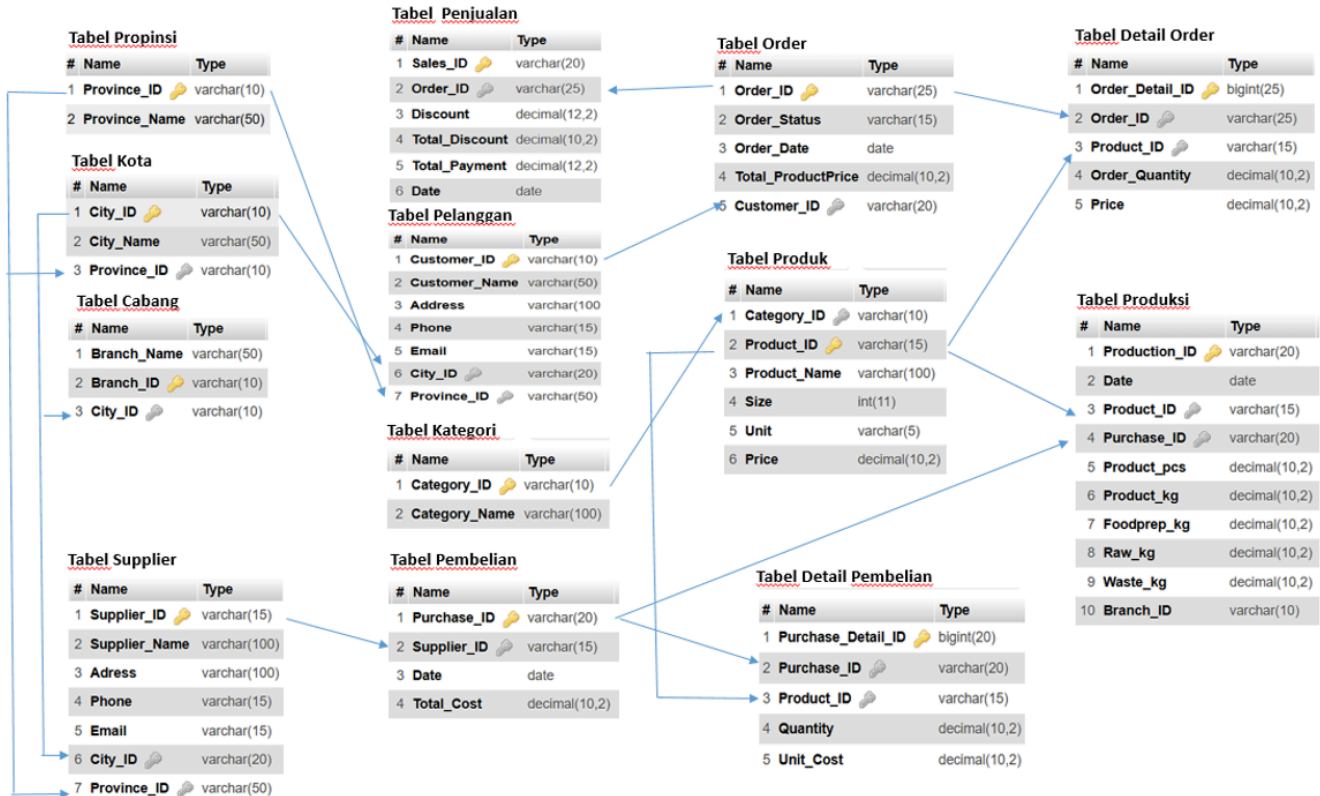
Pre-processing 1 --- > Filter, Split, Replace, Rename



Pre-processing 2 --- > Hapus Outlier --> 21.178 baris



5. Normalisasi databased dan design RDBMS (MySQL dan phpMyAdmin)



phpMyAdmin

Server: 127.0.0.1 Database: sayur_express

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers

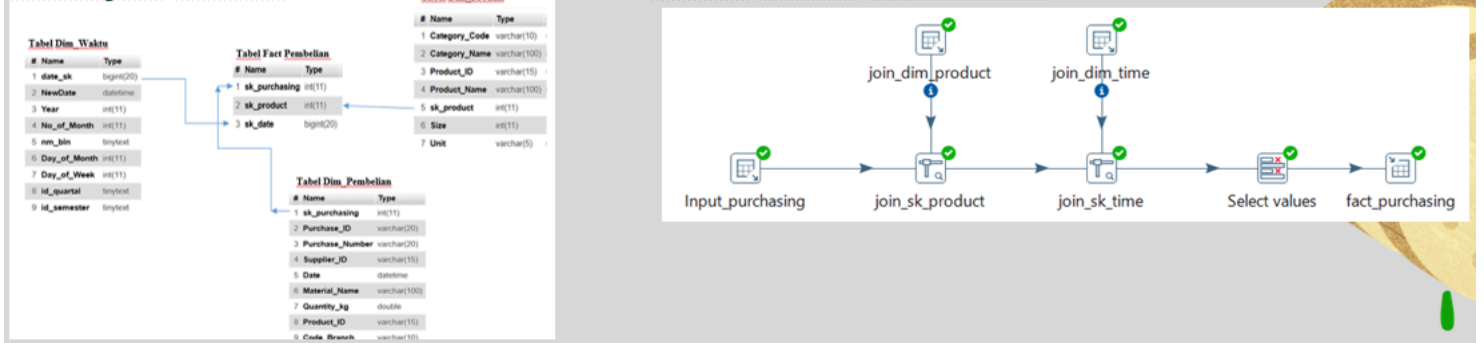
Filters

Containing the word:

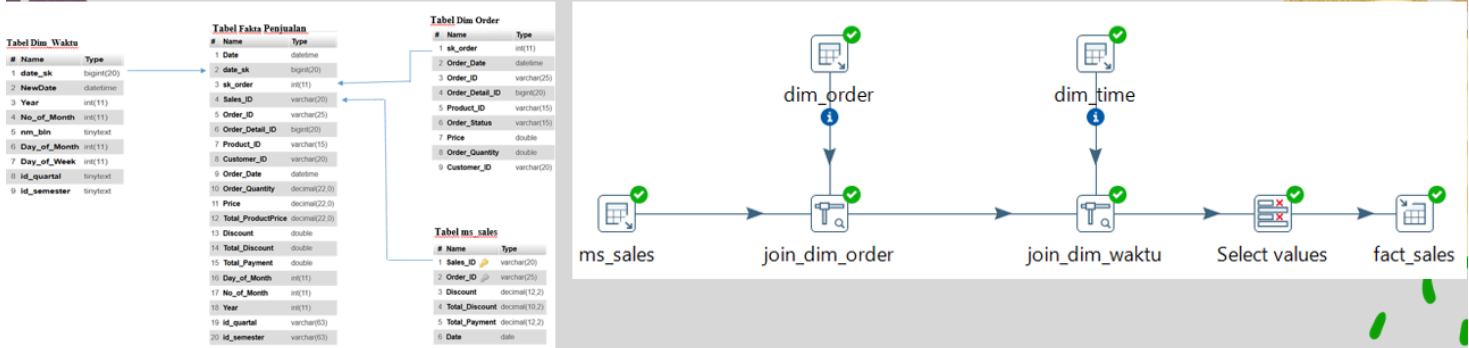
Table	Action	Rows	Type	Collation	Size	Overhead
☐ dim_branch	★ Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ dim_order	★ Browse Structure Search Insert Empty Drop	5,914	InnoDB	utf8mb4_general_ci	1.5 MiB	-
☐ dim_product	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ dim_production	★ Browse Structure Search Insert Empty Drop	180	InnoDB	utf8mb4_general_ci	80.0 KiB	-
☐ dim_purchasing	★ Browse Structure Search Insert Empty Drop	180	InnoDB	utf8mb4_general_ci	64.0 KiB	-
☐ dim_supplier	★ Browse Structure Search Insert Empty Drop	1	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ dim_time	★ Browse Structure Search Insert Empty Drop	3,600	InnoDB	utf8mb4_general_ci	272.0 KiB	-
☐ fact_penjualan	★ Browse Structure Search Insert Empty Drop	5,914	InnoDB	utf8mb4_general_ci	2.0 MiB	-
☐ fact_produk	★ Browse Structure Search Insert Empty Drop	180	InnoDB	utf8mb4_general_ci	80.0 KiB	-
☐ fact_proses	★ Browse Structure Search Insert Empty Drop	180	InnoDB	utf8mb4_general_ci	64.0 KiB	-
☐ fact_sisapersediaan	★ Browse Structure Search Insert Empty Drop	180	InnoDB	utf8mb4_general_ci	80.0 KiB	-
☐ fac_pembelian	★ Browse Structure Search Insert Empty Drop	180	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ ms_branch	★ Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ ms_category	★ Browse Structure Search Insert Empty Drop	5	InnoDB	utf8mb4_general_ci	16.0 KiB	-
☐ ms_city	★ Browse Structure Search Insert Empty Drop	7	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ ms_customer	★ Browse Structure Search Insert Empty Drop	12	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ ms_order	★ Browse Structure Search Insert Empty Drop	5,914	InnoDB	utf8mb4_general_ci	1.8 MiB	-
☐ ms_order_details	★ Browse Structure Search Insert Empty Drop	5,914	InnoDB	utf8mb4_general_ci	1.0 MiB	-
☐ ms_product	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ ms_production	★ Browse Structure Search Insert Empty Drop	180	InnoDB	utf8mb4_general_ci	80.0 KiB	-

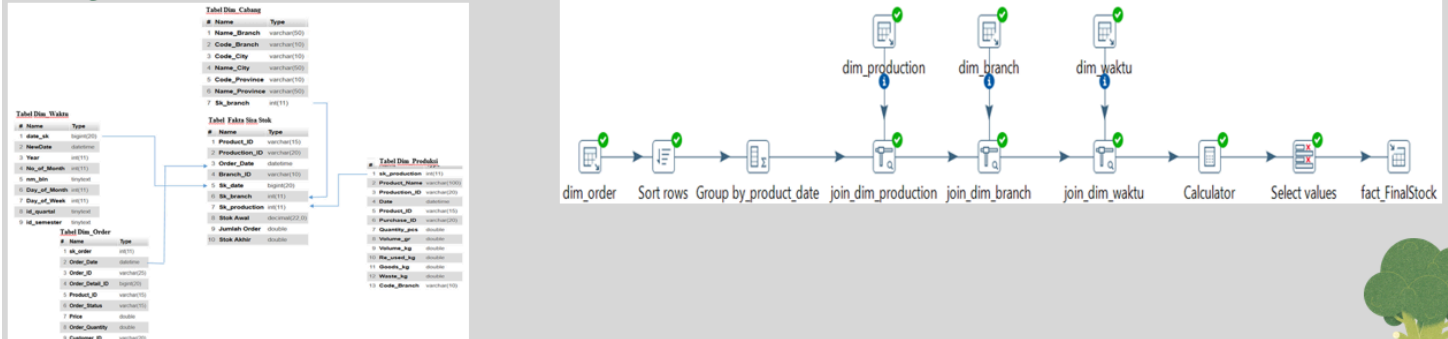
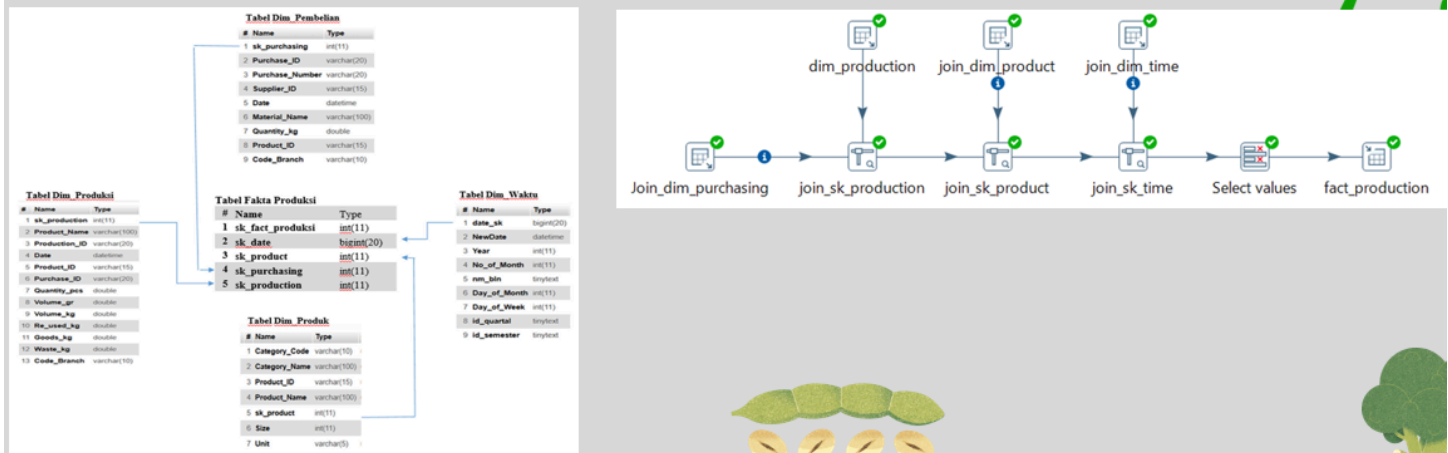
6. Proses ekstraksi dan transformasi data dengan Pentaho Data Intergration

Intelijen Bisnis -> ETL --> OLAP --- >Tabel Fakta Pembelian



Intelijen Bisnis -> ETL --> OLAP --- >Tabel Fakta Penjualan





7. Loading - visualisasi data dan informasi dengan Power BI

